

Luke Sarfas

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Technical Skills

Programming: Java, Python, Dart, Swift, C, Haskell, PHP, HTML/CSS/JS

Databases: DynamoDB, SQL, Firebase

Frameworks/Libraries: React.js, Node.js, Flutter

Miscellaneous: AWS, GitHub, VBA, Linux

Experience

Amazon Web Services (AWS) - *Software Development Engineer*

September 2024 – Present

- AWS CloudWatch Predictions team, working on the Anomaly Detection service, with a focus on building and optimising scalable systems for metric forecasting.
- Prototyped internal GenAI-powered developer tools to assist in debugging and incident triage
- Intimate knowledge of AWS services, developing and launching full-stack applications on AWS.
- Delivered support for cutting edge anomaly detection models, and delivering optimisations to reduce training latency by over 80%.
- Resolved critical DynamoDB throttling issues by implementing a caching strategy to track user behaviour, reducing database strain and improving system efficiency.
- Independently investigated cost-cutting mechanisms, resulting in the development of an automated system for detecting and deleting expired models, effectively eliminating redundant processing and achieving cost savings of over \$500,000 annually.
- Spearheaded the operational support campaign, automating the deployment of several pipelines via CI/CD, alongside ensuring security compliance with AWS and high-side environment standards.
- Built observability dashboards and incident response workflows, cutting MTTR to under 10 minutes.

Skills: AWS, Cloud Infrastructure, Scalable Service Architecture, Operations

Unidingo (Formerly Butterfly) - *Senior Software Engineer*

Sept 2022 – September 2024

- Developed a mobile social media application for both iOS and Android platforms using Flutter, integrating Firebase for real-time data handling and user authentication.
- Participated in a dynamic startup environment, rapidly adapting to evolving requirements and significantly contributing to the company's growth, from 100 to over 20,000 users during my employment.
- Routinely engaged with the user community, analysing and implementing feedback to refine application features, enhance user experience, and address potential issues promptly.
- Entrusted with leadership responsibilities, leading a team of 9.
- Initiated and authored comprehensive documentation and procedural guidelines, laying the groundwork for consistent coding practices.
- Actively participated in high-level strategic discussions, providing technical insights and recommendations to shape Butterfly's future product roadmap and business strategies.

Skills: Dart, Flutter, Firebase, UI/UX Design, Leadership, Agile Development

Deloitte - *IT Specialist Intern*

June 2023 – July 2023

- Engaged in comprehensive IT System Audits, Reviews, and Assessments to ensure alignment with organisational visions, missions, and goals.
- Assisted in the identification and analysis of IT-related risks, providing mitigation and removal strategies to uphold information system functionality.
- Participated in Vulnerability Assessments and Penetration Testing to identify IT vulnerabilities and recommend security enhancements.
- Presented for interviews with customers and IT staff, analysing documentation and software controls.

- Supported the assessment of information system adequacy, evaluating its relevance and support to the organisation's overall business operations.

Skills: VBA, Omnia, IT Audit, Problem Solving, Communication & Networking

Education

BSc (Hons) Computer Science, University of Warwick Sept 2021 - June 2024: - First Class (top 10%)

A Levels, Gower College Swansea 2019 – 2021:

5 A* (top 0.1%)- Computer Science, Mathematics, Further Mathematics, Music, EPQ

GCSEs, Bishopston Comprehensive School 2014 – 2019: Achievements: 10 A*, 2 A

Projects

C99 Compiler in C++

September 2023 – Present: Developed a compiler for C99, a simplified version of the C language. The compiler translates C99 code to LLVM intermediate representation using the LLVM 17.0.1 framework.

Skills: Compiler Design, Lexical Analysis, Syntax Parsing, C++, C

Music Performance Analysis Tool (Third Year Project)

July 2023 – Present: Developed software designed to aid musicians, especially beginners, in enhancing their skills. The tool offers real-time feedback on pitch and rhythm accuracy and generates tailored practice exercises based on a user's performance.

Skills: Audio Processing, JavaScript, React.js, TensorFlow.js, Web Audio API, Full Stack Development

Project Management System (Group Project)

January 2023 – March 2023: Developed a real-time project management system, which advised users on areas of the project which might cause development to fail. I was the project manager, for which I had to design the structure of the program, create the documentation, organise Sprint cycles, develop the algorithms used in the predictive analysis.

Skills: Agile, Team Collaboration, Java, Python, Neural Networks, Web Development

Timetable Scheduling Program

November 2022 – January 2023: Developed a task scheduler as a solution to a coursework. This involved implementing simulated annealing to quickly and effectively find an optimal solution.

Skills: Heuristics, User-defined Constraints, Java

Network Packet Sniffer

October 2022 – December 2022: Developed a robust packet sniffer in C, designed to intercept and analyze network traffic in real-time. The tool identifies packet types, sources, and also handles blacklisted URLs. Implemented a Threadpool model to manage high traffic loads efficiently, ensured unique IP address tracking and offers comprehensive testing capabilities for SYN and ARP packets as well as URL filtering.

Skills: C, Network Packet Analysis

They Said What? Sentiment analysis tool (Sponsored by ARM)

March 2020 – April 2020: Developed a natural language processor for sentiment analysis on tweets. This involved using the Twitter API to sample texts on a specific day, which would be processed to evaluate and quantify the polarity and subjectivity of the text. Applications of the tool explored include gauging public sentiment towards political figures and new legislation.

Skills: Natural Language Processing, Web Development, API Integration, Python, HTML, CSS, JavaScript

Miscellaneous

- **Cyber First Advanced Course:** Cyber Security course completed in July 2020
- **Freelance Writing:** Wrote articles on computer hardware for gaming PCs for geekawhat.com
- **Google Hash Code:** Participated in February 2022 in a team of 4

Personal Interests

Playing music (3 Grade 8 distinctions in Music Performance for Saxophone, Clarinet, Piano, and Grade 8 Music Theory) with my band and orchestra, Surfing, Martial Arts.

References available on request.